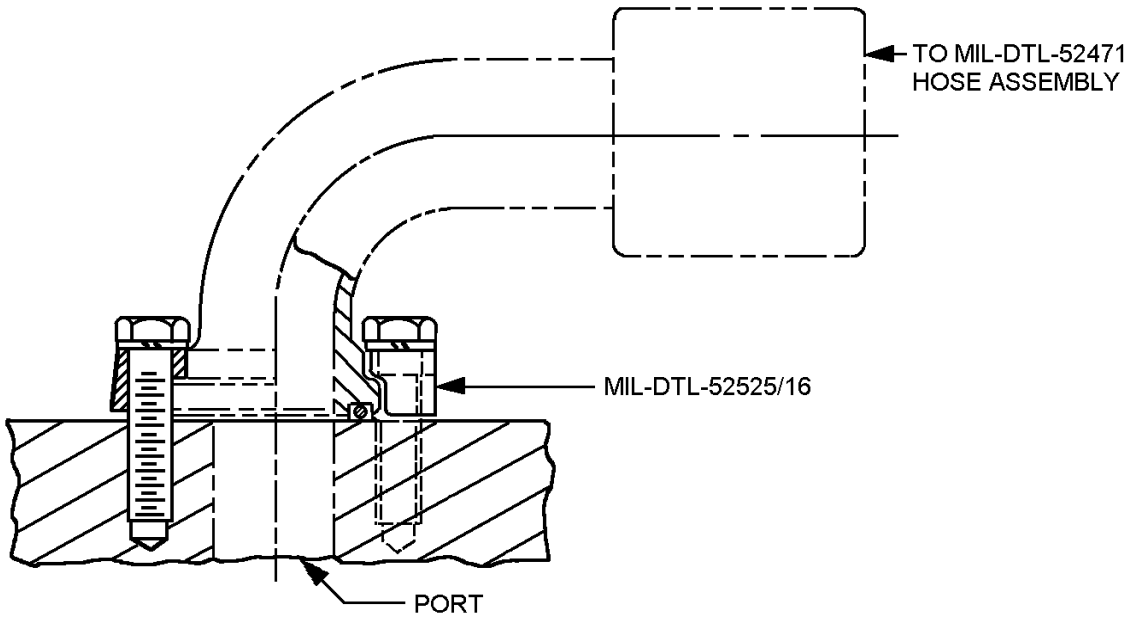
INCH-POUND
MIL-DTL-52525/17D
16 December 2019
SUPERSEDING
MIL-DTL-52525/17C
20 November 2008

DETAIL SPECIFICATION SHEET

# FITTINGS, CLAMP-HALVES, FLANGE TO PORT, FOR 4-BOLT SPLIT-FLANGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-52525.

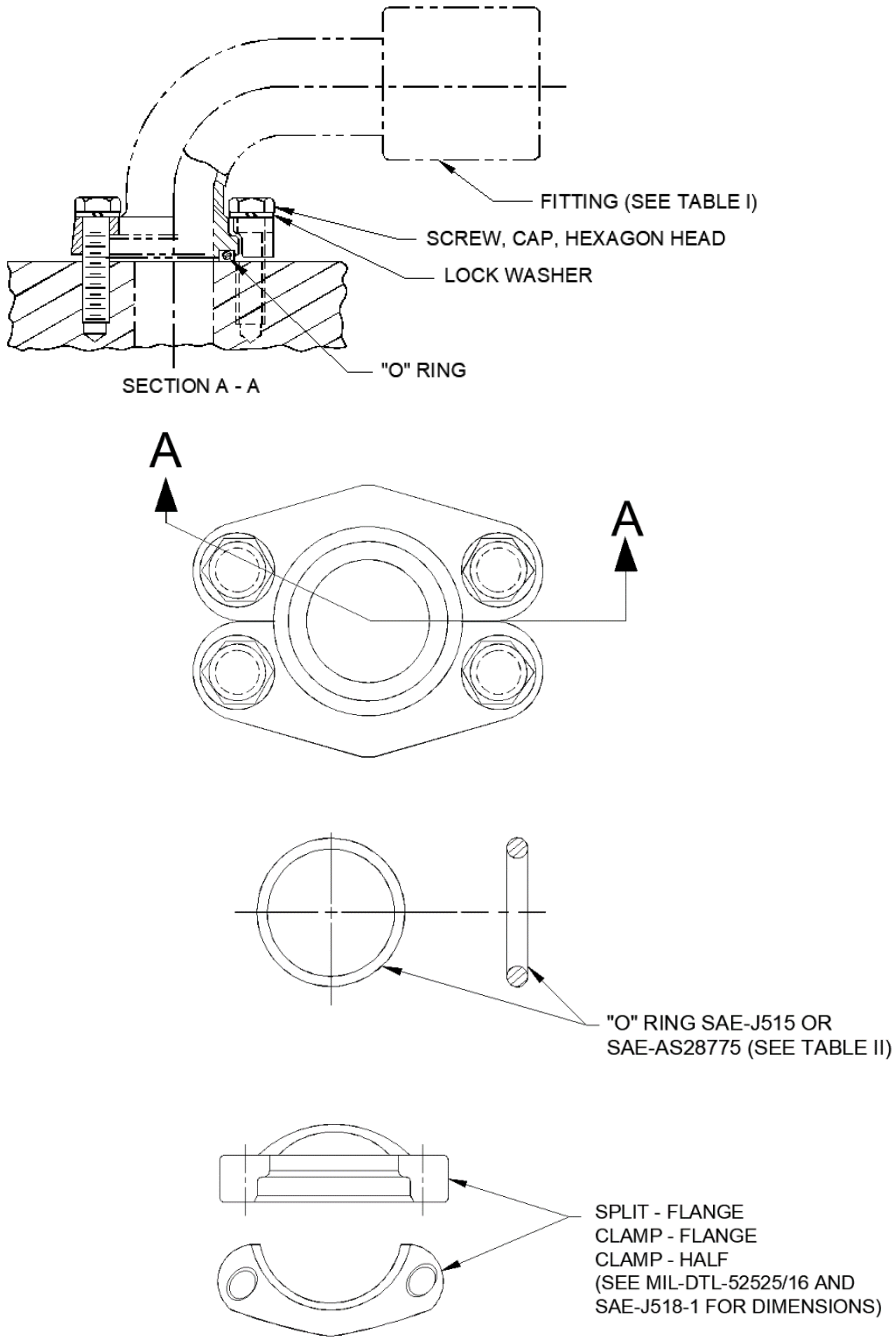



See note at end of figure.

FIGURE 1. Connections: flange to port, 4-bolt split-flange type.



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NOTE: The fitting clamp-half flange to port is MIL-DTL-52525/16. This connection is used with MIL-DTL-52471 hose assemblies.

FIGURE 1. Connections: flange to port, 4-bolt split-flange type - Continued.



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TABLE I. Connecting units used with flange type.

| Part or Identifying Number (PIN) | Fitting | Screw cap hex head (4 required) NASM90725 PIN 1/ 2/ | Washer, lock helical, split (4 required) NASM35338 PIN 1/ 2/ | Clamp-halves flange to port | Clamp-halves flange to port SAE-J518-1 PIN |
|---|---|---|---|---|---|
| M52525/17-8 | M52525/10-8-8 | NASM90725 -36 | MS35338-45 | M52525/16-8 | J518-8 1161 |
| | M52525/11-8-8 | | | | |
| | M52525/12-8-8 | | | | |
| M52525/17-12 | M52525/10-12-12 | NASM90725 -62 | MS35338-46 | M52525/16-12 | J518-12 1161 |
| | M52525/11-12-12 | | | | |
| | M52525/12-12-12 | | | | |
| | M52525/13-12-12 | | | | |
| | M52525/14-12-12 | | | | |
| | M52525/15-12-12 | | | | |
| M52525/17-16 | M52525/10-16-16 | NASM90725 -62 | MS35338-46 | M52525/16-16 | J518-16 1161 |
| | M52525/11-16-16 | | | | |
| | M52525/13-16-16 | | | | |
| | M52525/14-16-16 | | | | |
| | M52525/15-16-16 | | | | |
| | M52525/43-16-16 | | | | |
| | M52525/44-16-16 | | | | |
| | M52525/45-16-16 | | | | |
| M52525/17-20 | M52525/10-16-20 | NASM90725 -89 | MS35338-47 | M52525/16-20 | J518-20 1161 |
| | M52525/10-20-20 | | | | |
| | M52525/11-16-20 | | | | |
| | M52525/11-20-20 | | | | |
| | M52525/12-16-20 | | | | |
| | M52525/12-20-20 | | | | |
| | M52525/13-20-20 | | | | |
| | M52525/14-20-20 | | | | |
| | M52525/15-20-20 | | | | |
| | M52525/43-20-20 | | | | |
| | M52525/44-20-20 | | | | |
| | M52525/45-20-20 | | | | |
| M52525/17-24 | M52525/10-20-24 | NASM90725 -113 | MS35338-48 | M52525/16-24 | J518-24 1161 |
| | M52525/10-24-24 | | | | |
| | M52525/11-24-24 | | | | |
| | M52525/12-20-24 | | | | |
| | M52525/12-24-24 | | | | |
| | M52525/13-24-24 | | | | |
| | M52525/14-24-24 | | | | |
| | M52525/15-24-24 | | | | |
| | M52525/43-24-24 | | | | |
| | M52525/44-24-24 | | | | |

See footnotes at end of table.



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TABLE I. Connecting units used with flange type - Continued.

| PIN | Fitting | Screw cap hex head (4 required) NASM90725 PIN 1/ 2/ | Washer, lock helical, split (4 required) NASM35338 PIN 1/ 2/ | Clamp-halves flange to port | Clamp-halves flange to port SAE-J518-1 PIN |
|---|---|---|---|---|---|
| M52525/17-32 | M52525/10-24-32 | NASM90725 -113 | MS35338-48 | M52525/16-32 | J518-32 1161 |
| | M52525/10-32-32 | | | | |
| | M52525/11-32-32 | | | | |
| | M52525/12-32-32 | | | | |
| | M52525/43-32-32 | | | | |
| | M52525/44-32-32 | | | | |
| | M52525/45-24-24 | | | | |
| | M52525/45-32-32 | | | | |

1/ Bolts and washers identified in table I are for low carbon steel materials only. If another material listed in table III is used to manufacture bolts or washers the dimensional requirements shall be in accordance with NASM90725 and NASM35338.
2/ Never reuse fasteners after they have been subjected to loading or the elements.



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The O-ring legacy tables II and III shows O-rings originally called out by MIL-DTL-52525/17 and shows the superseding PIN for a side by side comparison.

TABLE II. O-ring legacy replacements for AN6227. 1/ 2/ 3/

| Dash size | PIN | Width (T or d2) Inches (mm) | ID (d1) Inches (mm) | Comments |
|---|---|---|---|---|
| -8 | AN6227-15 | .0934 (2.37) | .750 (19.05) | Originally MIL-DTL-52525/17 called out the AN6227 and AN6230 PIN's. AN6227 and AN6230 are cancelled and superseded by SAE-AS28775. |
| -12 | AN6227-19 | .0934 (2.37) | 1.000 (25.40) | |
| -16 | AN6227-24 | .125 (3.18) | 1.313 (33.35) | |
| -20 | AN6227-27 | .125 (3.18) | 1.500 (38.10) | |
| -24 | AN6230-3 | .125 (3.18) | 1.875 (47.63) | |
| -32 | AN6230-6 | .125 (3.18) | 2.250 (57.15) | |
| Replacement PIN for the AN6227 | | | | |
| Dash size | PIN | Width (T or d2) Inches (mm) | ID (d1) Inches (mm) | Comments |
| -8 | SAE-AS28775-210 | .135-.143 (3.43 - 3.63) | .728 - .0740 (18.49 - 1.880) | The SAE-AS28775 PIN is a replacement for AN6227 and AN6230. |
| -12 | SAE-AS28775-214 | .135-.143 (3.43 - 3.63) | .978 - .0990 (24.84 - 2.514) | |
| -16 | SAE-AS28775-219 | .135-.143 (3.43 - 3.63) | 1.290 - 1.302 (32.76 - 33.070) | |
| -20 | SAE-AS28775-222 | .135-.143 (3.43 - 3.63) | 1.478 - 1.490 37.54 - 37.846) | |
| -24 | SAE-AS28775-328 | .205-.215 (5.21 - 5.46) | 1.840 - 1.860 (46.74 - 47.24) | |
| -32 | SAE-AS28775-331 | .205-.215 (5.21 - 5.46) | 2.215 - 2.235 (56.26 - 56.77) | |

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ The width and ID columns are dimensions from the PIN in column 2.



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TABLE III. O-ring legacy replacements for SAE-J120. 1/ 2/ 3/

| Dash size | PIN | Width (T or d2) Inches (mm) | ID (d1) Inches (mm) | Comments |
|---|---|---|---|---|
| -8 | SAE-J120-210 | .139 ±.004 (3.53 ± 0.10) | .734 (18.64) | SAE-J120 is cancelled without replacement. However the SAE-J120 cancellation notice should state "superseded by SAE-J515". |
| -12 | SAE-J120-214 | .139 ±.004 (3.53 ± 0.10) | .984 (24.99) | |
| -16 | SAE-J120-219 | .139 ±.004 (3.53 ± 0.10) | 1.296 (32.92) | |
| -20 | SAE-J120-222 | .139 ±.004 (3.53 ± 0.10) | 1.484 (37.69) | |
| -24 | SAE-J120-225 | .139 ±.004 (3.53 ± 0.10) | 1.859 (47.22) | |
| -32 | SAE-J120-228 | .139 ±.004 (3.53 ± 0.10) | 2.234 (56.74) | |
| Replacement PIN for the SAE-J120 | | | | |
| Dash size | PIN | Width (T, or d2) Inches (mm) | ID (d1) Inches (mm) | Comments |
| -8 | SAE-J515-210 | .1389 ±.004 (3.528 ± 0.01) | .734 (18.64) | The SAE-J515 PIN is a replacement for SAE-J120 |
| -12 | SAE-J515-214 | .1389 ±.004 (3.53 ± 0.10) | .98 (24.89) | |
| -16 | SAE-J515-219 | .1389 ±.004 (3.53 ± 0.10) | 1.296 (32.92) | |
| -20 | SAE-J515-222 | .1389 ±.004 (3.53 ± 0.10) | 1.484 (37.69) | |
| -24 | SAE-J515-225 | .1389 ±.004 (3.53 ± 0.10) | 1.859 (47.22) | |
| -32 | SAE-J515-228 | .1389 ±.004 (3.53 ± 0.10) | 2.234 (56.74) | |

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ The width and ID columns are dimensions from the PIN in column 2.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and in tables I and IV.

The O- ring shall have a durometer hardness of 90 ±5.

If fittings are to be used on an oxygen hose, see MIL-DTL-52525.

Materials and finishes shall be in accordance with MIL-DTL-52525 and table I.

Finish. Finishes shall be in accordance with table I. All plating's shall be capable of meeting 96-hour salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.



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TABLE IV. Material and finish identification codes.

| PIN code material/plating finish | Material | Plating finish |
|---|---|---|
| A | Aluminum alloy 6061 or 7075 | Anodize in accordance with MIL-A-8625, type II. |
| AN | | Anodize in accordance with MIL-A-8625, type II and NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type II, class 1A. TCP Applied after anodizing. |
| BN | | Bare aluminum with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A. |
| C | Steel 2/ | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 8/ |
| YC | Chrome-molybdenum steel alloy 4130 | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 5/ 8/ |
| G | Steel 2/ | Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 25. 3/ 4/ |
| YG | Chrome-molybdenum steel alloy 4130 | |
| H | Steel 2/ | Zinc phosphate finish in accordance MIL-DTL-16232, type Z, class1. |
| YH | Chrome-molybdenum steel alloy 4130 | |
| J | Steel 2/ | Zinc plating with a colored chromate conversion in accordance with ASTM B633; type II, Fe/Zn 8, or ASTM B695, type II, class 8. 3/ 5/ |
| YJ | Chrome-molybdenum steel alloy 4130 | |
| K | Nickel-copper alloy UNS N04400 | No additional finish. |
| M | High-chromium nickel alloy UNS N06690 | No additional finish. |
| N | Steel 2/ | Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 4/ |
| YA | Chrome-molybdenum steel alloy 4130 | |
| YN | | Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 4/ 7/. |
| P | Steel 2/ | Zinc plating with colorless passivate in accordance with ASTM B633, type V, Fe/Zn 8. 3/ 4/ |
| YP | Chrome-molybdenum steel alloy 4130 | |
| S | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7. |
| T | Titanium | Anodize in accordance with SAE-AMS2488, type 2. 6/ 9/ |
| TF | | Fluoride phosphate in accordance with SAE-AMS2486. 6/ 9/ |
| U | Steel 2/ | Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. 4/ |
| V | | Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 4/ 7/. |
| W | Steel 2/ | Any zinc plating above. 3/ |
| YW | Chrome-molybdenum steel alloy 4130 | |
| Z | Steel | Zinc plating in accordance with ASTM B695, type II, class 8. 3/ 5/ |

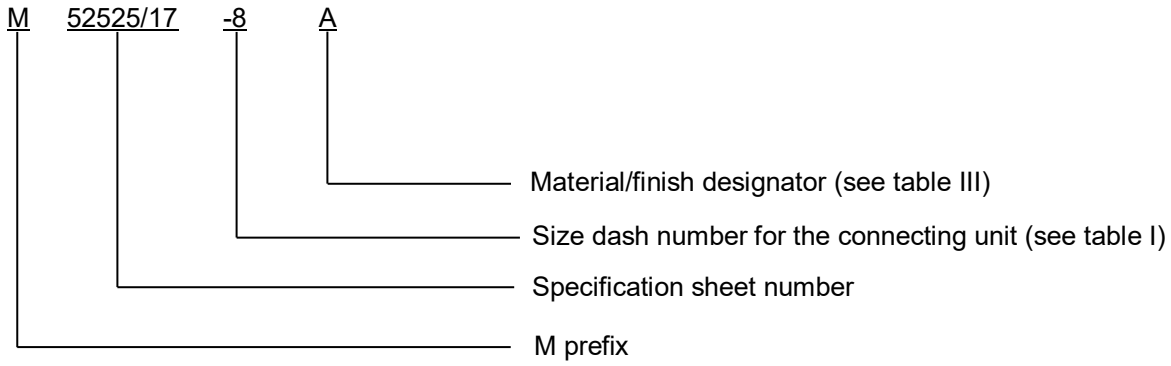
1/ Embrittlement test need not be run.
2/ All steels except 4130.
3/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.
4/ Hexavalent chromium free. Finish shall be RoHS compliant.
5/ These finishes have a chromate conversion coating or chromic acid rinse which includes hexavalent chromium and are not recommended. The use of coatings depositing hexavalent chromium for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.
6/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293.
7/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
8/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA. The use of cadmium plating for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.



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9/ Titanium shall not be used in oxygen or potable water systems.

PIN: The PIN consists of the letter “M” the specification sheet number, a dash, a number for the connecting unit, and a material finish designator.




PIN example: M52525/17-8A, describes a .500 inch (12.70 mm) 4-bolt split-flange, aluminum with an anodized finish.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when the other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

The use of cadmium plating or coatings depositing hexavalent chromium for Army vehicles is prohibited unless otherwise explicitly stated in the contract as a requirement.

Class I and II ozone depleting substances (ODS) shall not be used in MIL-DTL-52525/17 or any referenced procedures.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Color identification. Color identification shall be in accordance with SAE-AS4841.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.



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Referenced documents. In addition to MIL-DTL-52525, this document references the following:

| | | |
|---|---|---|
| MIL-A-8625 | SAE-AMS-STD-595/36081 | SAE-AMS-C-81562 |
| MIL-DTL-16232 | SAE-AMS-STD-595/36099 | SAE-AMS-QQ-P-416 |
| MIL-DTL-52471 | SAE-AMS-STD-595/36118 | SAE-AMS2417 |
| MIL-DTL-52471/16 | SAE-AMS-STD-595/36134 | SAE-AMS2488 |
| MIL-DTL-81706 | SAE-AMS-STD-595/36152 | SAE-AMS2486 |
| ASTM B117 | SAE-AMS-STD-595/36170 | SAE-AMS2700 |
| ASTM B633 | SAE-AMS-STD-595/36173 | SAE-AS4841 |
| ASTM B695 | SAE-AMS-STD-595/36176 | SAE-AS28775 |
| ASTM F1136/F1136M | SAE-AMS-STD-595/36231 | SAE-J515 |
| ASTM MNL37 | SAE-AMS-STD-595/36251 | SAE J518-1 |
| NASM 35338 | SAE-AMS-STD-595/36270 | T9070-AL-DPC-020/077-2 |
| NASM 90725 | SAE-AMS-STD-595/36280 | |
| SAE-AMS-STD-595/36076 | SAE-AMS-STD-595/36293 | |

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2020-002)

Review activities:
Army - CR4
Navy - AS, CG, SA, YD

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